Deterministic Distance-Based Source Sampling for Efficient Multi-Scale Network Centrality Analysis

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Abstract

Localised network centrality is routinely computed at multiple distance thresholds, from pedestrian to metropolitan scales. The analysis is particularly expensive at long distances, where each node's shortest-path traversal must explore a large reachable subgraph containing many thousands of nodes. Statistical sampling can yield large speedups, but the appropriate sampling strategy is not obvious for the localised setting.

We focus on a practical constraint common in comparative urban analysis: sampling probabilities should be deterministic and consistent across locations at the same analysis distance, so that centrality estimates remain directly comparable across different networks and morphologies. We use source-sampling with distance-bounded Brandes traversals, applying a single deterministic distance-only schedule $p(d)$ to both closeness and betweenness. The schedule is motivated by the Hoeffding bound, using a canonical grid model to convert analysis distance to a fixed sampling probability without requiring any network-specific information.

We empirically validate this approach on the Greater London (156,285 nodes) and Greater Madrid street networks at distances from 1–20 km. The deterministic schedule ($\varepsilon = 0.05$) yields strong rank agreement with ground truth (Spearman $\rho \geq 0.98$) and speedups of up to 11.0×, while remaining sufficiently conservative to preserve rank ordering across heterogeneous networks. The approach is implemented in the open-source `cityseer` Python package.

Keywords: network centrality, sampling, urban morphology, betweenness, harmonic closeness, computational efficiency

1 Introduction

Urban morphological analysis makes use of localised network centrality (typically variants of closeness or betweenness computed within distance thresholds) to reveal patterns of accessibility and flow in street networks. Because these metrics are typically evaluated at multiple scales, from pedestrian (∼500 m) to metropolitan (∼20 km), the cost is driven by the longest distances, where each node's traversal must explore a large reachable subgraph. For a network of 156,285 nodes, a single multi-scale centrality analysis can take hours.

Each source's traversal contributes to the centrality estimates of all nodes it reaches, so running traversals from a random fraction of sources and rescaling yields approximate estimates for the entire network. The typical number of reachable nodes — which we call *reach* r — varies with the distance threshold and governs both the cost and the effectiveness of this strategy. At short distances, reach is small, traversals are cheap, and there is little to gain from sampling. At long distances, reach is large and traversals are expensive, but each node's estimate also draws on contributions from many sources, making sampling both most beneficial and most reliable.

Sampling has been analysed for global centrality, but the localised setting changes the estimand: only pairs within the cutoff distance contribute. For closeness, the Eppstein–Wang

source-sampling bound (Eppstein and Wang, 2004) motivates replacing global node count n with mean reach r, giving $k = O(\log r/\varepsilon^2)$. For betweenness, path-sampling methods such as Riondato–Kornaropoulos (Riondato and Kornaropoulos, 2016) are theoretically attractive on global graphs, but under localisation each useful sample still incurs shortest-path work while informing only one path. By contrast, a single source-based bounded traversal with Brandes backpropagation informs all nodes in the local neighbourhood. This difference in per-traversal information yield explains why path-sampling became prohibitively expensive as we moved to smaller localisations, and why practical use would require a network-specific switch to exact computation. Both closeness and betweenness are therefore handled with one deterministic distance-only source-sampling schedule $p(d)$.

Our contributions are:

1. We construct a deterministic distance-only source-sampling schedule $p(d)$, motivated by the Hoeffding bound and a canonical grid reach model, that applies identically to both closeness and betweenness across all networks without requiring network-specific calibration.

2. We argue that path-sampling methods (Riondato–Kornaropoulos and related frameworks) lose their practical efficiency under localisation: each sampled path still requires shortest-path work but yields only one path update, so conservative budgets can cross over and become slower than exact localised computation at short distances.

3. We empirically validate this schedule on the Greater London (156,285 nodes) and Greater Madrid networks at distances from 1–20 km with $\varepsilon = 0.05$, demonstrating speedups of up to 11.0× while maintaining $\rho \geq 0.98$.

The approach is implemented in the open-source `cityseer` Python package.

Paper structure. Section 2 defines the centrality measures and experimental protocol. Section 3 constructs the deterministic distance-based sampling schedule, analyses its error structure on synthetic networks, and positions it relative to prior bounds. Section 4 reports empirical validation on two large street networks, followed by practical guidance and limitations in Section 5.

2 Methods

We consider two centrality measures, both computed with a distance cutoff $d_{\max}$ as is common in urban network analysis.

Harmonic closeness (Rochat, 2009; Freeman, 1979) measures how near a node is to the rest of its distance-bounded neighbourhood: nodes with many nearby destinations receive higher values. It sums inverse distances to all nodes within the threshold:

$$C_H(v;\, d_{\max}) = \sum_{\substack{u \neq v \\ d(v,u) \leq d_{\max}}} \frac{1}{d(v,u)} \tag{1}$$

Betweenness (Freeman, 1977) measures how often a node lies on shortest routes between other nodes, capturing potential "through-movement" within the distance-bounded neighbourhood. It counts the fraction of shortest paths (with total length at most $d_{\max}$) that pass through each node:

$$C_B(v;\, d_{\max}) = \sum_{\substack{s \neq t,\ s \neq v,\ t \neq v \\ d(s,t) \leq d_{\max}}} \frac{\sigma_{st}(v)}{\sigma_{st}} \tag{2}$$

where σ_{st} is the number of shortest paths from s to t, and $\sigma_{st}(v)$ is the number passing through v. Common distance thresholds range from 400 m to 20 km.

A quantity central to the sampling analysis is *reach*: the mean number of nodes within $d_{\max}$ (including the source node itself),

$$r = \frac{1}{N} \sum_{v} |\{u : d(v, u) \leq d_{\max}\}| \tag{3}$$

In a canonical grid network with inter-node spacing $s = 175\,\text{m}$, reach grows from approximately 100 nodes at 1 km to over 40,000 at 20 km. This order-of-magnitude variation across distance thresholds directly governs the relationship between sample size and accuracy.

Synthetic networks. We generate synthetic networks using 3 topology templates: trellis (high connectivity, representing planned urban cores), tree (low connectivity, representing suburban sprawl), and linear (intermediate connectivity). These span the range of urban street network structure. Each topology is tiled to create networks of approximately 12 km extent; we evaluate accuracy on a central 4 km×4 km window to reduce edge effects. We test 4 distance thresholds from 250 m to 4,000 m, and 3 error tolerances (ε) from 0.01 to 0.5.

Boundary effects. Nodes near network edges have artificially lower centrality because their catchments are truncated. We address this by applying an inward buffer from the convex hull of all nodes, equal to the maximum analysis distance (4 km for synthetic networks, 20 km for validation). Only nodes inside this buffered zone are marked as "live" and included in accuracy calculations, ensuring all live nodes have complete catchments at all tested distances. This approach is consistent with common practice for localised network analysis (Gil, 2017).

Accuracy metrics. We measure sampling accuracy using two complementary metrics: Spearman rank correlation and normalised additive error.

Spearman rank correlation (ρ) between sampled and ground-truth centrality values:

$$\rho = 1 - \frac{6 \sum_i (R_i^{\text{sampled}} - R_i^{\text{true}})^2}{N(N^2 - 1)} \tag{4}$$

where R_i denotes the rank of node i. Spearman correlation is appropriate because many urban applications primarily use *relative* rankings (e.g., identifying the most central streets) rather than absolute values. Source-sampling also tends to estimate high-centrality nodes more precisely because they receive contributions from many sampled sources, which helps preserve the top of the ranking.

Normalised additive error $\varepsilon_{\text{obs}} = \max_v |C^{\text{sampled}}(v) - C^{\text{true}}(v)| \,/\, C_{\max}$, where $C_{\max}$ is an upper bound on the centrality value within the distance-bounded neighbourhood: $r(r-1)$ for betweenness and r for harmonic closeness. This normalised error lies in $[0, 1]$ and corresponds to the ε parameter used to set the sampling schedule.

Validation networks. We validate the model on two real-world street networks. The primary validation uses the Greater London Authority (GLA) road network, sourced from Ordnance Survey Open Roads data. After preprocessing (removing filler nodes, consolidating nearby nodes, and removing dangles), the network comprises approximately 156,285 nodes. We apply a 20 km inward buffer to eliminate edge artifacts. As an external validation, we also test the model on the Greater Madrid metropolitan street network following the same protocol.

3 A Unified Sampling Model and Deterministic Distance Schedule

Figure 1 illustrates why sampling-based centrality is attractive. For both closeness (A) and betweenness (B), accuracy (Spearman ρ) improves as ε decreases (more samples), with the curves flattening well before the smallest ε values. The shaded bands show 95% confidence intervals across topologies and distances. The remainder of this section constructs the deterministic source-sampling schedule, motivated by the Hoeffding bound, and explains why path-sampling

frameworks, while theoretically appealing for betweenness on global graphs, do not transfer to the localised setting.

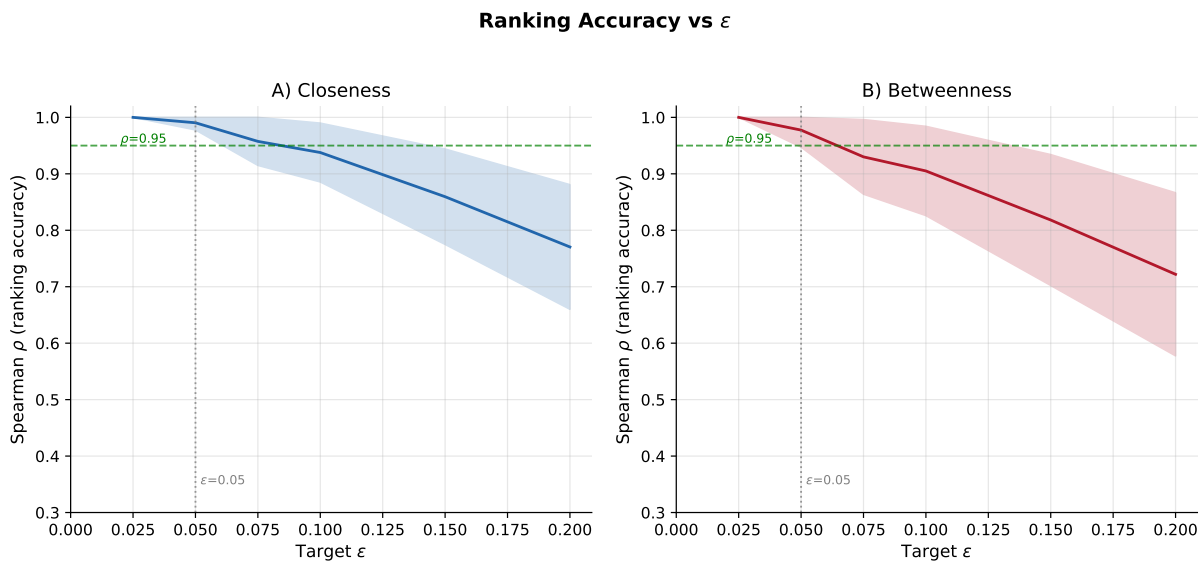

Figure 1: **The sampling opportunity.** Ranking accuracy (Spearman ρ) vs. error tolerance ε for closeness (A) and betweenness (B), averaged across synthetic topologies and distances. The green dashed line marks $\rho = 0.95$; the grey dotted line marks the paper default $\varepsilon = 0.05$. Shaded bands show 95% confidence intervals.

From reach bounds to a deterministic schedule.

Closeness: Hoeffding source-sampling. The Eppstein–Wang framework (Eppstein and Wang, 2004) provides the starting point for closeness. When k sources are sampled uniformly, each contributes a value in $[0, 1]$ (after normalisation) to each target node's closeness estimate. By Hoeffding's inequality (Hoeffding, 1963), the average of k such bounded contributions is concentrated around its expectation; a union bound extends this guarantee to all target nodes simultaneously. In the original global setting the union bound runs over all n nodes, yielding sample complexity $k = \log(2n/\delta)/(2\varepsilon^2)$.

For localised centrality with distance threshold $d_{\max}$, only nodes within a distance-bounded neighbourhood can receive non-zero contribution from a given sampled source. This motivates replacing the global factor n with the mean reach r, giving

$$k = \frac{\log(2r/\delta)}{2\varepsilon^2} \tag{5}$$

with corresponding sampling probability

$$p = \min\left(1,\ \frac{k}{r}\right) \tag{6}$$

This model has no fitted parameters: ε (error tolerance) and δ (failure probability) are user-chosen conventions. At default values $\varepsilon = 0.05$, $\delta = 0.1$:

- **Low reach** ($r \lesssim 1{,}600$): $k > r$, so $p = 1$ and computation is exact (the schedule saturates naturally).
- **High reach**: p decreases as $O(\log r/r)$, so speedup $1/p$ grows as $O(r/\log r)$.

From reach-based to deterministic sampling probability. Equations 5–6 express p as a function of the mean reach r, which varies across distance thresholds and across networks. Applying the bound directly from observed reach — computing r by a pilot survey of the specific network at the specific distance, then deriving p — produces different sampling probabilities for different networks and locations at the same nominal distance. This is undesirable in localised centrality analysis, where a primary goal is to compare nodes across different locations (e.g., two districts of the same city, or two cities with different morphology). If the sampling probability is reach-dependent and varies across locations, the resulting estimates are not directly comparable: a node in a dense urban core receives a lower p than a node in a sparse suburb, even at the same analysis distance, introducing differential precision that confounds cross-location ranking.

A deterministic schedule that depends only on the distance threshold d eliminates this confound. Following a canonical grid network model with inter-node spacing s, reachability can be approximated as $r \approx \pi d^2/s^2$, from which the Hoeffding bound gives a closed-form $p(d)$ that is identical for all networks and locations at a given d. Using a fixed grid spacing $s = 175\,\mathrm{m}$ (a conservative estimate of the inter-node spacing of a sparse street network) produces a schedule that is deterministic, parameter-free given ε and δ, and sufficiently conservative: it *underestimates* reach for dense networks, yielding p values that are slightly higher than strictly necessary, but guarantees comparable precision across all locations. This is the schedule implemented in `cityseer`.

Betweenness: why path-sampling does not transfer, and why source-sampling does. Betweenness contributions are sparser and heavier-tailed than closeness contributions: most sources route zero shortest paths through a given node, while a few high-centrality nodes contribute heavily. Path-sampling methods such as Riondato–Kornaropoulos (Riondato and Kornaropoulos, 2016) exploit this structure on *global* graphs by sampling individual shortest paths between random node pairs, providing betweenness guarantees with

$$m = O\left(\frac{\log \mathrm{VD} + \log(1/\delta)}{\varepsilon^2}\right) \tag{7}$$

sampled paths. Their practical appeal is that point-to-point shortest-path search can terminate once t is settled, often before building a full shortest-path tree.

Under a localisation cutoff $d_{\max}$, this advantage weakens for two reasons. First, the admissible pair set depends on network distance (and therefore on local morphology), so there is no fixed global pair space from which to sample cheaply. Second, one path sample still incurs point-to-point shortest-path work yet updates betweenness only along one path, while one source traversal with Brandes backpropagation updates all nodes in the bounded neighbourhood. In street-like networks this creates a strong amortisation gap: per traversal, source-sampling typically yields far more node-level dependency information than path-sampling.

A simple cost model clarifies the switch problem. Let $m(d)$ be the path-sample budget at distance d, let $c_{\mathrm{pair}}(d)$ be expected cost per admissible path sample, and let $c_{\mathrm{src}}(d)$ be cost per bounded source traversal with Brandes backpropagation. Then

$$T_{\mathrm{path}}(d) \approx m(d)\, c_{\mathrm{pair}}(d), \qquad T_{\mathrm{exact}}(d) \approx n\, c_{\mathrm{src}}(d). \tag{8}$$

The approximate method is slower than exact when

$$m(d) \geq m^\star(d) := n\, \frac{c_{\mathrm{src}}(d)}{c_{\mathrm{pair}}(d)}. \tag{9}$$

As d decreases, conservative budgets needed for stable local rankings can increase while per-sample information decreases, pushing $m(d)$ toward $m^\star(d)$. Because both cost terms depend on morphology, $m^\star(d)$ is not a portable function of distance alone. This is why we found

no robust distance-only threshold for switching RK-style approximation to exact computation across different networks.

Source-sampling resolves the betweenness problem without additional cost. A single distance-bounded Dijkstra traversal from a sampled source s identifies all reachable nodes, and via Brandes backpropagation accumulates betweenness path counts for all r reached nodes in one pass. By analogy with the closeness case, we apply the same Hoeffding-motivated schedule to betweenness, giving a single schedule for both metrics:

$$k = \frac{\log(2r/\delta)}{2\varepsilon^2}, \qquad p = \min\left(1,\ \frac{k}{r}\right) \tag{10}$$

The cost per sampled source is higher for betweenness than closeness (Brandes backpropagation adds $O(r)$ work per source), but the same sampling probability is used for both. We validate empirically that this shared schedule is sufficiently conservative for both metrics (Section 4).

The logarithmic growth of k with r means that sampling requirements increase slowly with distance, producing speedups that grow with analysis scale. Table 1 provides the deterministic sampling probabilities at standard analysis distances for three ε values.

Table 1: Deterministic sampling schedule at standard analysis distances under different error tolerances ($\delta = 0.1$, $s = 175\,\mathrm{m}$). Paper default: $\varepsilon = 0.05$ for both metrics.

Distance	$\varepsilon = 0.05$		$\varepsilon = 0.1$		$\varepsilon = 0.2$	
	k	p	k	p	k	p
500 m	exact		exact		exact	
1 km	exact		exact		95	92.9%
2 km	exact		exact		112	27.5%
5 km	2,169	84.6%	542	21.1%	135	5.3%
10 km	2,446	23.8%	611	6.0%	152	1.5%
20 km	2,723	6.6%	680	1.7%	170	0.4%

Error structure across reachability. Although the additive error analyses target a uniform error criterion, it is natural to ask how accuracy varies across the reachability distribution. Figure 2 plots per-node errors against per-node reach, pooled across reach quartiles at distances where sampling occurs ($p < 1$) on the Greater London and Greater Madrid networks. A crossover holds for both closeness and betweenness: absolute error increases with reach, while normalised error decreases. This pattern has a natural interpretation. High-reach nodes sit at the centres of large local subgraphs; their centrality values are large sums over many contributions, so absolute estimation errors accumulate. But these same nodes enjoy high effective sample sizes, so the errors are small relative to the centrality magnitude. Conversely, low-reach peripheral nodes have small absolute errors but larger relative errors. In practical terms, high-centrality arterials and junctions tend to receive higher precision, a property we term *precision scales with importance*.

This error structure is consistent with the strong rank preservation observed empirically. The nodes with appreciable normalised error tend to fall at intermediate reach, where centrality values are comparatively clustered and small rank swaps contribute little to Spearman ρ. Meanwhile, the high-reach nodes whose ranking matters most receive the highest precision, and low-reach peripheral nodes approach exact computation.

Relationship to prior bounds. Table 2 places our approach in the context of the existing sampling literature.

All of these methods share a common pattern: sample a subset of computations, aggregate contributions at each node, and bound the resulting error. They differ in the unit of computation.

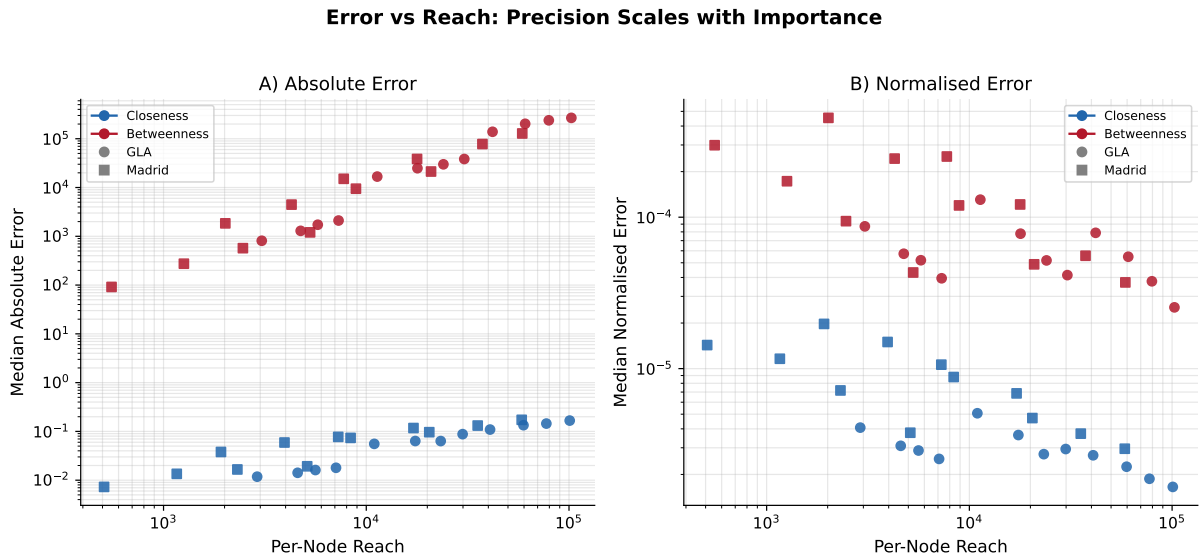

Figure 2: **Error crossover: precision scales with importance.** Per-node median absolute errors pooled across reach quartiles at distances where sampling occurs, for Greater London (circles) and Greater Madrid (squares). (A) Absolute error increases with reach: high-centrality nodes accumulate larger absolute errors because their centrality values are large sums over many contributions. (B) Normalised error (divided by r for closeness, $r(r-1)$ for betweenness) decreases with reach: relative error shrinks because high-reach nodes enjoy higher effective sample sizes.

Table 2: Comparison of sampling approaches for centrality approximation. All prior methods target global centrality (no distance cutoff) on general graphs. Our approach uses a Hoeffding-motivated schedule with reach r in place of global node count n. The same schedule is applied to both closeness and betweenness via source-sampling with Brandes backpropagation; the extension to betweenness uses the same bound by analogy and is validated empirically.

Method	**Metric**	**Guarantee**	**Sample complexity**	**Key parameter**
Eppstein & Wang (2004)	Closeness	Additive, all nodes	$O(\log n \,/\, \varepsilon^2)$	Node count n
Bader et al. (2007)	Betweenness	Relative, per vertex	$O(1 \,/\, \varepsilon^2 b(v))$	Betweenness $b(v)$
Riondato & Korn. (2016)	Betweenness	Additive, all nodes	$O((\log \mathrm{VD} + \log(1/\delta)) \,/\, \varepsilon^2)$	Vertex diam. VD
KADABRA (2019)	Betweenness	Additive, all nodes	Adaptive	Diameter D
SILVAN (2023)	Betweenness	Non-uniform	Variance-dependent	Path variance
This work	**Both**	**Additive, all nodes**	$\boldsymbol{O(\log r \,/\, \varepsilon^2)}$	**Reach** r

Source-sampling methods (Eppstein–Wang, Bader et al., Brandes and Pich (Brandes and Pich, 2007)) run a full shortest-path tree from each sampled node, accumulating contributions at every reached target; Cohen et al. (Cohen et al., 2014) apply the same source-to-target pattern for global closeness, combining it with pivot-based structural estimation. Path-sampling methods (Riondato and Kornaropoulos, 2016; Borassi and Natale, 2019; Pellegrina and Vandin, 2023) instead sample individual shortest paths between random node pairs, incrementing betweenness estimates along each path.

For localised centrality, path-sampling methods (Riondato–Kornaropoulos, KADABRA, SILVAN) lose their per-sample cost advantage: each sampled path still requires shortest-path work but yields only one path-level update, whereas source-sampling with Brandes backpropagation yields neighbourhood-wide updates from the same traversal. Equation 9 formalises the practical crossover at which conservative path budgets become slower than exact localised computation, and why that crossover is morphology-dependent. Applying the Eppstein–Wang bound with n replaced by reach r provides one unified additive schedule for both metrics without requiring a separate switching rule.

Empirically, on the street networks studied here, the unified schedule at $\varepsilon = 0.05$ is conservative: observed normalised errors (Figure 2B) are substantially smaller than the Hoeffding worst-case bound of $\varepsilon = 0.05$, confirming that the default is a pragmatic choice when rank preservation is the primary objective.

Algorithm. Algorithm 1 summarises the complete procedure.

Algorithm 1 Deterministic Per-Distance Sampling for Network Centrality

Require: Network $G = (V, E)$, distance thresholds $D = \{d_1, \ldots, d_m\}$
Require: Error tolerance ε (default 0.05), confidence δ (default 0.1)
Require: Canonical grid spacing s (default 175 m)
Ensure: Closeness estimates $\hat{C}_H[d]$ and betweenness estimates $\hat{C}_B[d]$ for each $d \in D$

```
for each distance d ∈ D do
    Compute canonical reach: r_d ← πd²/s²
    Compute required sources: k_d ← log(2r_d/δ)/(2ε²)
    Compute sampling probability: p_d ← min(1, k_d/r_d)
    Sample sources S_d ⊆ V with probability p_d
    for each source s ∈ S_d do
        Run distance-bounded Dijkstra from s; accumulate closeness via inverse distances
        Run Brandes backpropagation; accumulate betweenness path counts
    end for
    Scale both estimates by 1/p_d
end for
return {Ĉ_H[d], Ĉ_B[d] : d ∈ D}
```

The sampling probability p_d depends only on the distance threshold d and the user-chosen parameters ε and δ. No network-specific information (such as actual reach) is required. The canonical grid spacing $s = 175\,\mathrm{m}$ is a conservative estimate of inter-node spacing for a sparse street network; denser networks will have higher actual reach, making the schedule conservative (it samples more than strictly necessary).

4 Validation

The sampling schedule is calibrated to additive error ε, but urban applications typically care about rank order. We therefore report Spearman ρ as the primary validation metric. The error structure observed in Section 3 suggests that the schedule should be conservative enough to

preserve rankings, particularly for high-centrality nodes; we test this empirically on two large street networks.

We evaluate on the GLA and Greater Madrid networks described in Section 2. Both closeness and betweenness are computed via source-sampling at the deterministic distance-based probability $p(d)$ (Algorithm 1), with $\varepsilon = 0.05$ and $\delta = 0.1$. For both networks, each sampled configuration is repeated with three random seeds and the mean reported to assess stability. Ground-truth centrality for the GLA network is computed from all sources ($p = 1.0$).

Greater London. Table 3 presents validation results for the Greater London network at five distances from 1–20 km. At all distances, both harmonic closeness and betweenness achieve $\rho > 0.95$, with the minimum observed $\rho = 0.9812$ (betweenness, 20 km). At short distances (1–4 km), the model prescribes $p = 1.0$ (exact computation). At longer distances, the prescribed sampling probabilities decrease with distance; meaningful wall-clock speedup emerges at 10 km (3.7×) and grows to 11.0× at 20 km.

Table 3: Sampling validation on Greater London network ($\varepsilon = 0.05$, $\delta = 0.1$, $s = 175$ m).

Dist.	p	ρ_c	**Spd**$_c$	ρ_b	**Spd**$_b$
1 km	100.0%	1.0000	1.0×	1.0000	1.0×
2 km	100.0%	1.0000	1.0×	1.0000	1.0×
5 km	84.6%	0.9988	1.0×	0.9994	1.1×
10 km	23.8%	0.9958	3.7×	0.9921	4.8×
20 km	6.6%	0.9949	11.0×	0.9812	15.1×

Network: Greater London Area. Deterministic distance-based schedule: same p for both metrics at each distance. Subscripts: c = closeness, b = betweenness.

Betweenness is consistently the binding constraint: at every distance, $\rho_{\text{betw}} < \rho_{\text{close}}$. This is expected. Betweenness contributions are sparser and heavier-tailed than closeness contributions: most sources route zero paths through a given node, so the effective per-node sample size is lower even at the same sampling probability. Both metrics nonetheless exceed $\rho \geq 0.95$ at all distances.

Greater Madrid. To assess transfer beyond London, we applied the identical protocol to the Greater Madrid metropolitan street network. Table 4 presents the results. The model achieves $\rho > 0.95$ for both closeness and betweenness at all tested distances. The minimum observed $\rho = 0.9810$ (betweenness, 20 km). Meaningful wall-clock speedup emerges at 10 km (3.7×) and grows to 12.6× at 20 km.

Table 4: Sampling validation on Greater Madrid network ($\varepsilon = 0.05$, $\delta = 0.1$, $s = 175$ m).

Dist.	p	ρ_c	**Spd**$_c$	ρ_b	**Spd**$_b$
1 km	100.0%	1.0000	1.0×	1.0000	1.0×
2 km	100.0%	1.0000	1.0×	1.0000	1.0×
5 km	84.6%	0.9990	1.2×	0.9993	1.2×
10 km	23.8%	0.9931	3.7×	0.9921	4.3×
20 km	6.6%	0.9900	12.6×	0.9810	14.9×

Network: Greater Madrid, 181,176 nodes. Deterministic distance-based schedule: same p for both metrics at each distance.

Figure 3 shows Spearman ρ vs. distance for both networks and both metrics. Both GLA and Madrid maintain $\rho \geq 0.95$ across all distances, with betweenness consistently lower than closeness. Figure 4 shows the corresponding wall-clock speedups: at short distances ($\leq$5 km) the sampling overhead exceeds the small reduction in sources, so no net speedup is observed; at 10 km and beyond, substantial gains emerge, reaching up to 11.0× for GLA at 20 km.

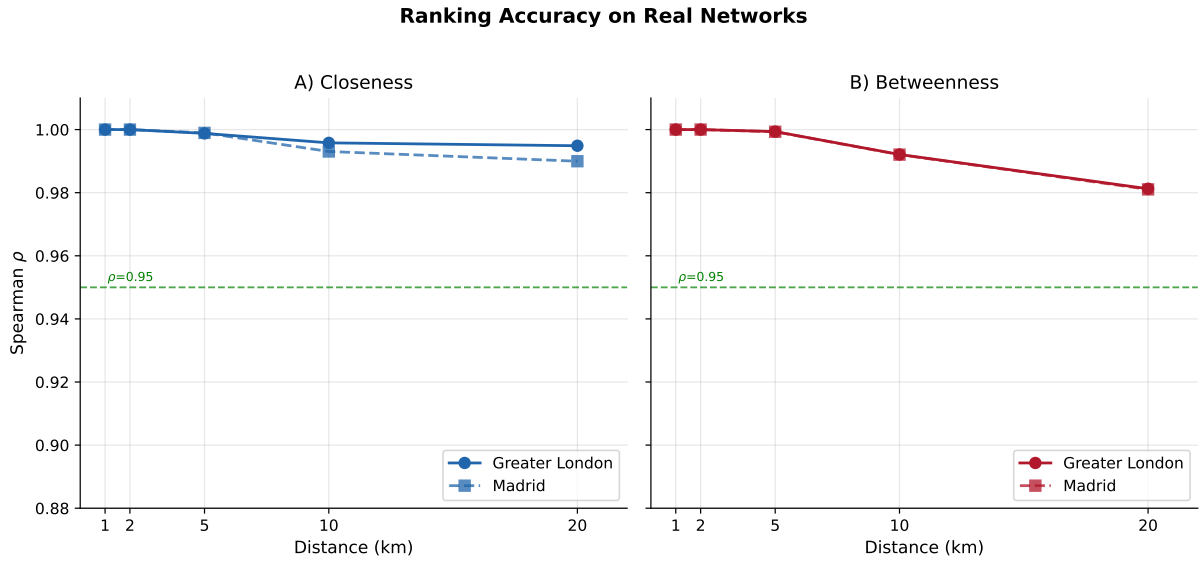

Figure 3: **Ranking accuracy on real networks.** Spearman ρ vs. analysis distance for Greater London (circles, solid) and Greater Madrid (squares, dashed), for closeness (A) and betweenness (B). The green dashed line marks $\rho = 0.95$. Both networks exceed the target at all distances; betweenness is the binding constraint.

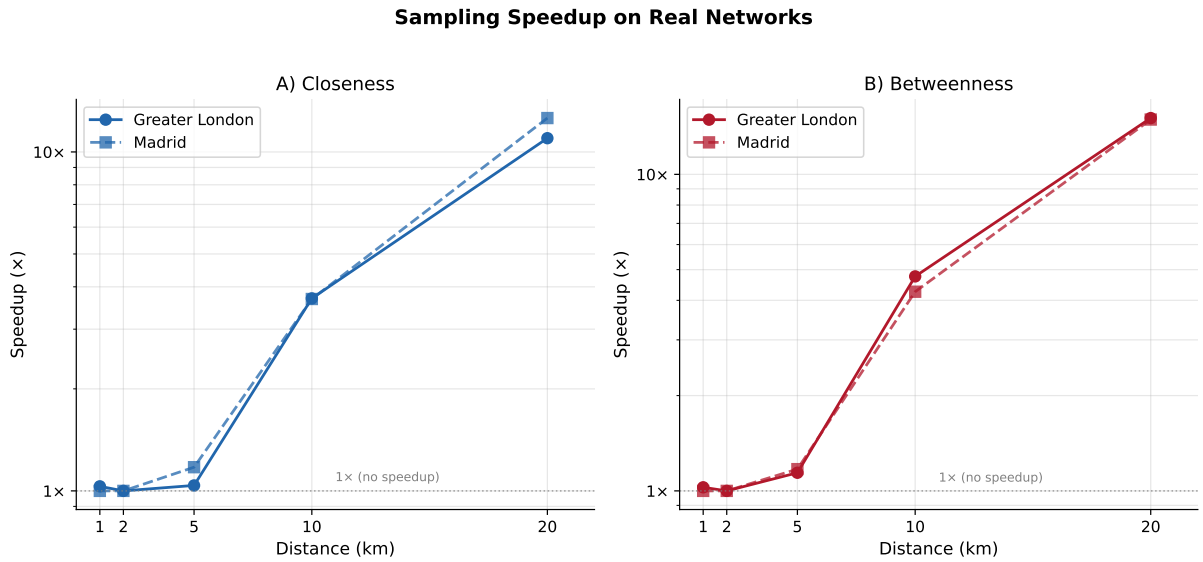

Figure 4: **Measured wall-clock speedup on real networks.** Speedup (baseline time / sampled time, log scale) vs. analysis distance for Greater London (circles, solid) and Greater Madrid (squares, dashed), for closeness (A) and betweenness (B). At short distances where p is close to 1, sampling overhead dominates and no wall-clock speedup is observed; meaningful speedup emerges at 10 km and beyond. At long distances, GLA achieves higher measured speedup primarily because the exact baseline traversals are costlier on the denser network; the deterministic schedule remains conservative for GLA (Figure 5).

Figure 5 plots mean reach vs. distance for both networks alongside the canonical model. GLA consistently exceeds the canonical curve (the schedule is conservative for GLA). Madrid falls below at longer distances, yet validation results confirm $\rho \geq 0.95$ is still achieved.

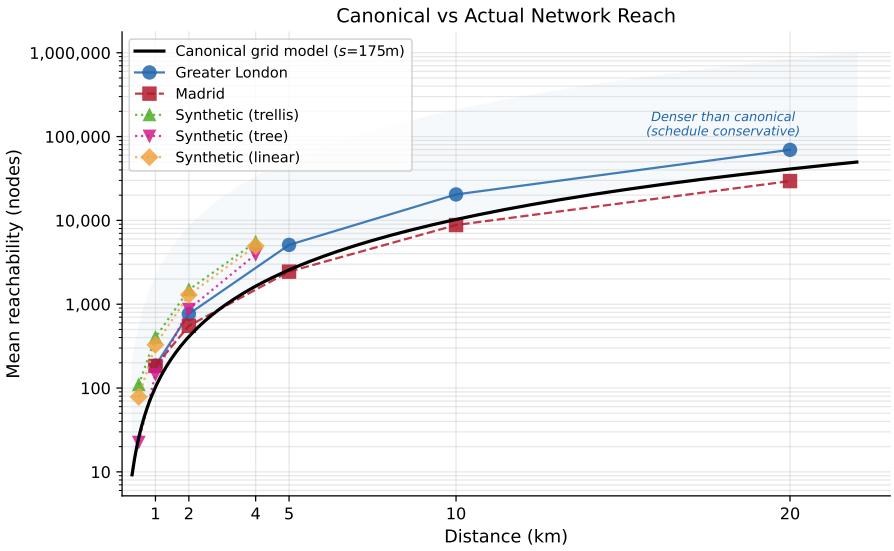

Figure 5: **Canonical vs. actual network reach.** Mean reachability (log scale) vs. analysis distance for Greater London, Greater Madrid, and three synthetic topologies, compared against the canonical grid model ($s = 175$ m, black line). Networks above the canonical curve are denser than assumed and the deterministic schedule is conservative. Networks below are sparser, yet both achieve $\rho \geq 0.95$ at all distances.

5 Discussion and Conclusion

Practical guidance. For users of the `cityseer` library, sampling is applied automatically: the user sets ε and δ, and the library derives the deterministic sampling probability $p(d)$ for each distance threshold and applies it uniformly to both closeness and betweenness via source-sampling. For manual implementations, follow Algorithm 1. Table 5 gives the deterministic schedule at standard analysis distances as a direct lookup. Figure 6 provides visual charts of the same schedule.

Table 5: Deterministic sampling schedule by analysis distance ($\varepsilon = 0.05$, $\delta = 0.1$, canonical grid spacing $s = 175$ m). Both closeness and betweenness use the same schedule. Sampling is exact ($p = 100\%$) for distances up to 4 km. Speedup shown is theoretical ($1/p$); measured wall-clock speedup is lower at near-unity p due to sampling overhead and emerges at 10 km and beyond.

Distance	**Canonical reach**	k	p (%)	**Speedup** ($1/p$)
1 km	103	1,525	100	1×
2 km	410	1,803	100	1×
5 km	2,565	2,169	84.6	1.2×
10 km	10,258	2,446	23.8	4.2×
20 km	41,033	2,724	6.6	15.1×

As a simple heuristic: at distances up to ∼4 km, $p = 1.0$ (exact computation). At 5–10 km, sample ∼25–85% of sources. At 20 km, sample ∼7%. Meaningful wall-clock speedup emerges at 10 km and beyond, where reduced sampling dominates framework overhead. Users requiring

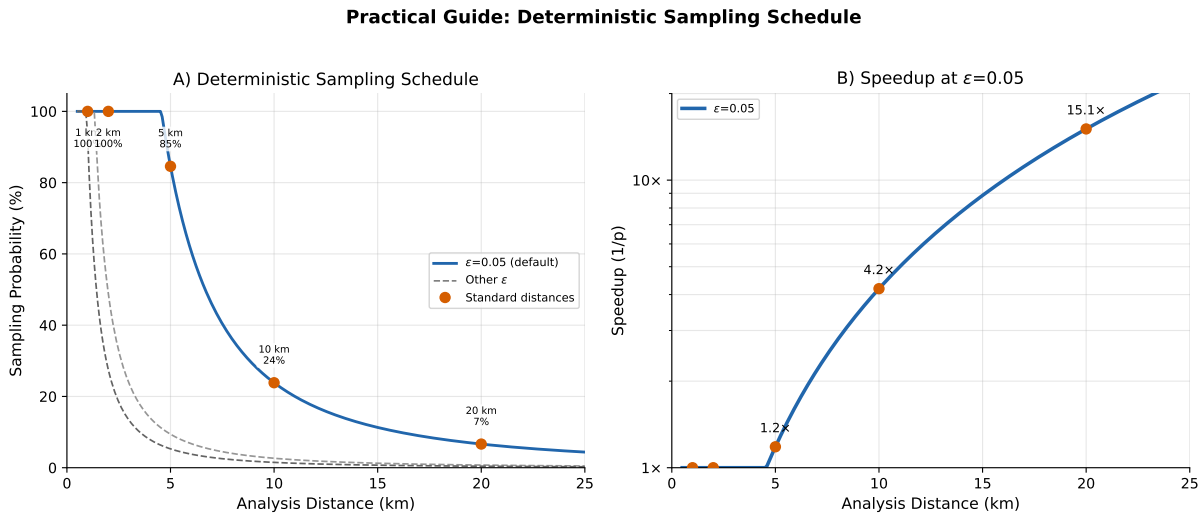

Figure 6: **Practical guidance.** (A) Deterministic sampling probability p vs. analysis distance under the canonical grid model ($\varepsilon = 0.05$, $\delta = 0.1$), with standard analysis distances annotated. (B) Theoretical speedup $1/p$ vs. distance; measured wall-clock speedup is lower at near-unity p due to sampling overhead (see Figure 4).

higher accuracy can reduce ε below the default 0.05 (halving ε approximately quadruples the required sources) or average 2–3 replications with different random seeds.

Sampling is not appropriate when exact centrality values (not approximations) are required, when centrality is needed for specific nodes rather than the full network, or when the network is small enough ($<$ 1,000 nodes) that full computation is already fast.

Limitations. Our approach rests on several assumptions. First, we assume uniform random sampling of source nodes. Non-uniform strategies (e.g., stratification by location or weighting toward high-centrality sources) could potentially achieve higher accuracy with fewer samples, but they would require proxies for node importance that are typically unavailable before computation.

Second, the deterministic schedule substitutes canonical grid reach $r = \pi d^2/s^2$ for the actual network reach; if a network is substantially sparser than the canonical model, the schedule under-samples relative to what the Hoeffding bound would prescribe from true reach. Validation on Madrid, which falls slightly below the canonical curve at longer distances, suggests this is tolerable in practice. Third, the model treats the network as statistically homogeneous: accuracy depends on distance rather than local structure. Validation on the GLA and Madrid networks, which include dense urban cores and sparse suburbs, suggests the schedule is conservative overall, but accuracy may vary locally. Relatedly, our analysis of RK-style path sampling indicates that localisation-specific switch points to exact computation are morphology-dependent (Equation 9); we therefore avoid hybrid switching rules and use one deterministic distance-only source schedule throughout.

Fourth, our experiments used shortest-path distance; angular (simplest-path) distance, common in space syntax analysis, may exhibit different accuracy characteristics and warrants separate validation. Fifth, validation is limited to two European cities; networks with substantially different morphology (e.g., North American grids, informal settlements) would strengthen external validity. Finally, betweenness shows thinner margins than closeness at high reach: the minimum observed $\rho = 0.9810$ (Madrid betweenness, 20 km) remains above the 0.95 threshold but leaves less headroom than closeness.

Future work. Several extensions could build on this work. Adaptive per-node sampling, where sampling probability varies with local reach, could improve efficiency in heterogeneous networks. Importance sampling (weighting toward influential sources) could reduce variance.

Online stopping criteria, as in KADABRA (Borassi and Natale, 2019), could replace a priori calibration with adaptive termination based on observed convergence; adapting such methods to localised, distance-bounded centrality would require new theoretical development. Systematic validation for angular distance, and extension to other centrality measures (eigenvector centrality, PageRank), would broaden applicability. Finally, variance-dependent analyses such as SILVAN (Pellegrina and Vandin, 2023), which replace worst-case combinatorial quantities with the observed estimator variance, are a promising direction for tightening localised bounds on well-structured network classes.

A simple deterministic source-sampling schedule yields strong rank preservation ($\rho \geq 0.98$) for localised centrality on the street networks studied here, at $\varepsilon = 0.05$ and without fitted parameters. Conceptually, this avoids the localisation mismatch of global path-sampling: under distance cutoffs, conservative path budgets can cross a morphology-dependent runtime switch point and become slower than exact computation (Equation 9). The error structure favours high-centrality nodes: high-reach nodes receive higher effective sample sizes and correspondingly more precise estimates, while low-reach peripheral nodes approach exact computation. Across the Greater London (156,285 nodes) and Greater Madrid networks at 1–20 km, speedups reach up to 11.0×, with all tested configurations exceeding $\rho \geq 0.95$. For city-scale networks where multi-scale centrality analysis previously required hours, this makes exploratory workflows and finer parameter sweeps practical.

The model is implemented in the open-source `cityseer` Python library.[1] Analysis scripts and validation data are available in the repository's `analysis/sampling/` directory.

References

Bader, D. A., Kintali, S., Madduri, K., and Mihail, M. (2007). Approximating Betweenness Centrality. In *Algorithms and Models for the Web-Graph (WAW 2007)*, volume 4863 of *Lecture Notes in Computer Science*, pages 124–137. Springer.

Borassi, M. and Natale, E. (2019). KADABRA is an ADaptive Algorithm for Betweenness via Random Approximation. In *27th Annual European Symposium on Algorithms (ESA 2019)*, pages 20:1–20:18.

Brandes, U. and Pich, C. (2007). Centrality Estimation in Large Networks. In *Proceedings of the 33rd International Workshop on Graph-Theoretic Concepts in Computer Science (WG 2007)*, volume 4769 of *Lecture Notes in Computer Science*, pages 90–100. Springer.

Cochran, W. G. (1977). *Sampling Techniques*. Wiley, 3rd edition.

Cohen, E., Delling, D., Pajor, T., and Werneck, R. F. (2014). Computing Classic Closeness Centrality, at Scale. In *Proceedings of the Second ACM Conference on Online Social Networks (COSN)*, pages 37–50.

Eppstein, D. and Wang, J. (2004). Fast Approximation of Centrality. *Journal of Graph Algorithms and Applications*, 8(1):27–34.

Freeman, L. C. (1977). A Set of Measures of Centrality Based on Betweenness. *Sociometry*, 40(1):35–41.

Freeman, L. C. (1979). Centrality in Social Networks: Conceptual Clarification. *Social Networks*, 1(3):215–239.

[1] github.com/benchmark-urbanism/cityseer-api

Gil, J. (2017). Street network analysis "edge effects": Examining the sensitivity of centrality measures to boundary conditions. *Environment and Planning B: Urban Analytics and City Science*, 44(5):819–836.

Hagberg, A. A., Schult, D. A., and Swart, P. J. (2008). Exploring Network Structure, Dynamics, and Function using NetworkX. In *Proceedings of the 7th Python in Science Conference (SciPy 2008)*, pages 11–15.

Hájek, J. (1971). Comment on "An Essay on the Logical Foundations of Survey Sampling, Part One". *Metrika*, 17(1):1–30.

Hoeffding, W. (1963). Probability Inequalities for Sums of Bounded Random Variables. *Journal of the American Statistical Association*, 58(301):13–30.

Horvitz, D. G. and Thompson, D. J. (1952). A Generalization of Sampling Without Replacement from a Finite Universe. *Journal of the American Statistical Association*, 47(260):663–685.

Pellegrina, L. and Vandin, F. (2023). SILVAN: Estimating Betweenness Centralities with Progressive Sampling and Non-uniform Rademacher Bounds. *ACM Transactions on Knowledge Discovery from Data*, 18(3):1–40.

Riondato, M. and Kornaropoulos, E. M. (2016). Fast Approximation of Betweenness Centrality through Sampling. *Data Mining and Knowledge Discovery*, 30(2):438–475.

Rochat, Y. (2009). Closeness Centrality Extended to Unconnected Graphs: The Harmonic Centrality Index. In *Applications of Social Network Analysis (ASNA 2009)*.

Särndal, C.-E., Swensson, B., and Wretman, J. (2003). *Model Assisted Survey Sampling*. Springer, 2nd edition.

A Technical Details

Synthetic network generation. Synthetic networks are generated using tile-based templates. Each template defines a connectivity pattern within a single tile; tiles are then arrayed to create networks of the desired extent.

Trellis A regular grid with diagonal connections, representing highly connected urban cores. Average degree ≈ 4.

Tree A hierarchical branching structure where each node connects to at most 3 others. Represents suburban cul-de-sac development. Average degree ≈ 2.3.

Linear A constrained grid with fewer cross-connections than trellis. Intermediate connectivity. Average degree ≈ 3.

Networks are tiled 24×24 to achieve approximately 12 km extent, then the largest connected component is extracted. Node positions are perturbed slightly to avoid perfectly regular spacing.

GLA network preprocessing. The Greater London Authority road network is sourced from Ordnance Survey Open Roads data (GeoPackage format).

Preprocessing steps:

1. Remove invalid and empty geometries
2. Explode multi-part geometries

3. Build NetworkX graph from edges
4. Remove filler nodes (degree-2 nodes that don't represent junctions)
5. Remove dangling nodes (dead-end segments $< 20\,\mathrm{m}$)
6. Consolidate nearby nodes (threshold $1\,\mathrm{m}$)
7. Extract largest connected component

Final network: $\approx$ 156,285 nodes.

Correctness verification against NetworkX. To ensure the correctness of our Rust-based centrality implementation in `cityseer`, we conducted systematic verification against the widely-used NetworkX library (Hagberg et al., 2008). For each synthetic network topology and distance threshold combination, we computed both harmonic closeness and betweenness centrality using our Rust implementation (via `cityseer` Python bindings) and NetworkX's reference functions.

For each node v in the test networks, we computed the absolute difference

$$\Delta_v = |C_{\text{cityseer}}(v) - C_{\text{NetworkX}}(v)| \tag{11}$$

The maximum difference across all nodes and all test configurations was $\max_v \Delta_v < 10^{-6}$, attributable to floating-point arithmetic precision.

Verification was performed across all three synthetic topologies, distance thresholds from $200\,\mathrm{m}$ to $3200\,\mathrm{m}$, both shortest-path and angular distance heuristics, and both harmonic closeness and betweenness centrality.

Horvitz–Thompson estimator. When estimating population totals from samples, two classical approaches are the Horvitz–Thompson (H–T) estimator and the Hájek ratio estimator. Under Bernoulli (independent) source sampling with inclusion probabilities π_i, the Horvitz–Thompson estimator (Horvitz and Thompson, 1952) estimates a population total as $\hat{T}_{HT} = \sum_{i \in S} y_i/\pi_i$, where S is the set of sampled sources. This inverse-probability weighting yields an unbiased estimator whenever $\pi_i > 0$ for all sampled units.

The Hájek estimator (Hájek, 1971) instead uses a ratio form,

$$\hat{T}_H = N \cdot \frac{\sum_{i \in S} y_i/\pi_i}{\sum_{i \in S} 1/\pi_i}, \tag{12}$$

which is often lower-variance but can be slightly biased in finite samples (Cochran, 1977; Särndal et al., 2003). In `cityseer`, we use Horvitz–Thompson-style inverse-probability weighting because it matches the paper's $1/p$ rescaling description and extends directly to unequal inclusion probabilities (e.g., when combining a base sampling probability with per-source importance weights).

Implementation in `cityseer`. The sampling model is implemented in the `cityseer` Python library (version 4.23+). Both closeness and betweenness use the same deterministic distance-based Hoeffding schedule. Sampling is enabled via the `sample=True` flag:

```
from cityseer.metrics import networks

gdf = networks.closeness_shortest(
    network_structure,
    nodes_gdf,
    distances=[500,1000,2000,5000,10000],
```

```
    sample=True
)

gdf = networks.betweenness_shortest(
    network_structure,
    nodes_gdf,
    distances=[500,1000,2000,5000,10000],
    sample=True
)
```

The default error tolerance ($\varepsilon = 0.05$) and confidence ($\delta = 0.1$) are set in `cityseer.config`. Helper functions for custom workflows are available in `cityseer.config`, including:

- `compute_hoeffding_p` — Hoeffding bound from mean reach
- `compute_distance_p` — deterministic p from distance via canonical grid model

See the library documentation for details.

Reproducibility. All analysis scripts and validation results are available at analysis/sampling/. The current script pipeline runs sequentially from `00_generate_cache.py` through `05_generate_macros.py` via `run_all.py`, with each script's outputs documented in its header.